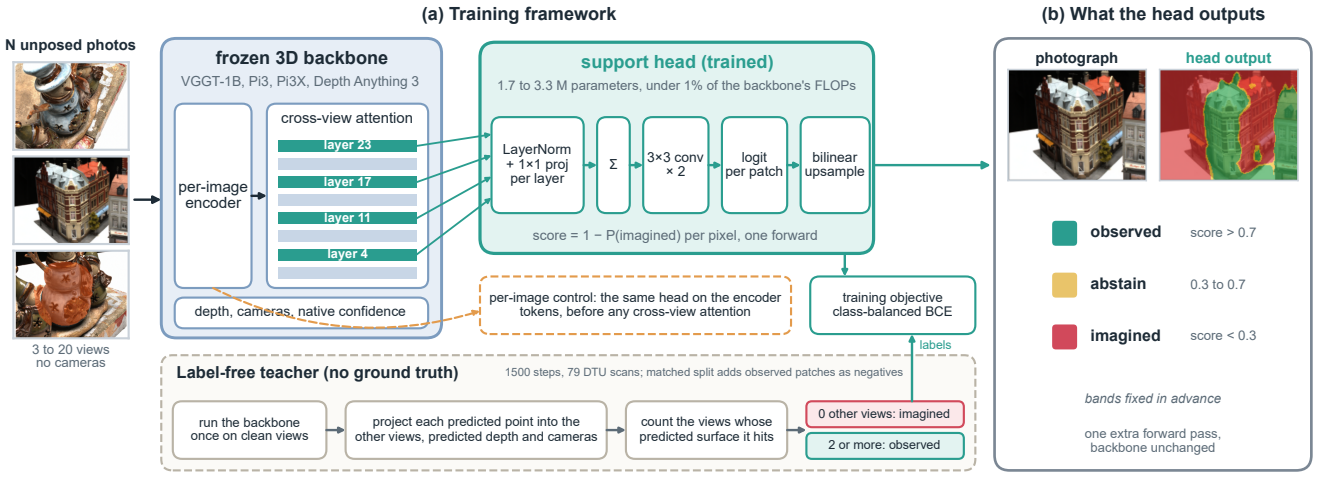

(a) Training framework
(b) What the head outputs
N unposed photos
3 to 20 views
no cameras
frozen 3D backbone
VGGT-1B, Pi3, Pi3X, Depth Anything 3
per-image encoder
cross-view attention
layer 23
layer 17
layer 11
layer 4
depth, cameras, native confidence
support head (trained)
1.7 to 3.3 M parameters, under 1% of the backbone's FLOPs
LayerNorm + 1×1 proj per layer
Σ
3×3 conv × 2
logit per patch
bilinear upsample
score = 1 − P(imagined) per pixel, one forward
per-image control: the same head on the encoder tokens, before any cross-view attention
training objective class-balanced BCE
labels
Label-free teacher (no ground truth)
1500 steps, 79 DTU scans; matched split adds observed patches as negatives
run the backbone once on clean views
project each predicted point into the other views, predicted depth and cameras
count the views whose predicted surface it hits
0 other views: imagined
2 or more: observed
photograph
head output
observed
score > 0.7
abstain
0.3 to 0.7
imagined
score < 0.3
bands fixed in advance
one extra forward pass, backbone unchanged